STAT 615
Group Project

Uzoma Ogechukwu Kanu
Ana Paula Toledo de Miranda
Jin Sook Song

Mediation Analysis

The Role of Educational Attainment in Residential Mobility Patterns Across Income Brackets

1. Introduction

This project investigates mediation analysis as a valuable methodological tool for understanding the complex relationships between socioeconomic factors and residential patterns. The study centers on exploring the degree to which education's impact on residential mobility is explained by income. The core question guiding this analysis is, "Does educational attainment affect residential mobility through income?" The underlying concept is that higher levels of education typically lead to improved employment opportunities and, consequently, increased income, which then influences an individual's capacity and inclination to relocate.

By exploring this potential mediating relationship, the research seeks to uncover the interconnected nature of social and economic forces in shaping residential choices and mobility patterns. This inquiry is particularly relevant for understanding broader trends in socioeconomic mobility, housing stability, and spatial inequality within the United States. Understanding these dynamics is crucial for policymakers and researchers aiming to address disparities and promote equitable opportunities for all individuals, regardless of their socioeconomic background.

This research acknowledges that while income is undoubtedly a critical factor, it may not be the sole pathway through which education impacts residential decisions. The goal is to provide a nuanced perspective on how education shapes not only economic prospects but also broader life circumstances that affect where people choose to live.

To conduct this analysis, data from the 2023 U.S. Census American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) Dataset is utilized. A random sample of 50,000 records has been extracted from the larger dataset, balancing manageability with representativeness. The analysis aims to capture the indirect effect of educational attainment on mobility through income, acknowledging that other factors beyond income may also play a role. By examining this pathway, the research seeks to contribute to a richer understanding of how education and income, together, shape patterns of residential stability and change across the United States.

2. Methods

2.1. Requirements and Assumptions of Mediation Analysis

Mediation analysis is a statistical approach used to understand the mechanism through which an independent variable (X) influences a dependent variable (Y) via a mediator variable (M). For credible results, several key requirements and assumptions must be met:

- Causal Order: The independent variable should precede the mediator, which in turn should precede the outcome variable.
- No Omitted Confounders: There should be no unmeasured confounders affecting the relationships between X and M, or M and Y.
- Linearity and Additivity: The relationships among variables are typically assumed to be linear and additive, unless otherwise specified.
- Measurement Reliability: All variables should be measured accurately and reliably.
- Sufficient Sample Size: Adequate sample size is necessary to detect mediation effects with sufficient statistical power.
- Correct Model Specification: The model should include all relevant variables and correctly specify the functional forms of relationships.

2.2. Benefits of Mediation Analysis

Mediation analysis offers several important benefits in research:

- Mechanistic Insight: It helps uncover the processes or pathways through which an effect occurs, providing a deeper understanding beyond simple associations.
- Theory Testing: Mediation analysis allows researchers to test theoretical models about how and why variables are related.
- Intervention Development: By identifying mediators, researchers can target interventions more effectively to influence outcomes.
- Partitioning Effects: It distinguishes between direct and indirect effects, showing the relative importance of different pathways.

2.3. Cautions and Limitations of Mediation Analysis

Despite its benefits, mediation analysis has several cautions and limitations:

- Causal Inference Challenges: Establishing causality is difficult, especially in non-experimental or observational studies.
- Assumption Violations: Results can be biased if key assumptions (e.g., no omitted confounders) are violated.
- Measurement Error: Inaccurate measurement of the mediator or other variables can distort findings.
- Temporal Ambiguity: If the temporal order of variables is unclear, mediation claims may be unwarranted.
- Complexity with Multiple Mediators: Models with multiple mediators or non-linear relationships require advanced methods and careful interpretation.
- Overinterpretation: There is a risk of overinterpreting indirect effects as evidence of true mediation when alternative explanations exist.

3. Example Research Question

This project focuses on mediation analysis as the central methodological approach, both as a subject of study and as a practical analytical tool. To contextualize this methodological focus, a sample research question is posed: "Does educational attainment affect residential mobility through income?" This study investigates the role of educational attainment in shaping residential mobility patterns across different income brackets. The central research question guiding this analysis is: Does educational attainment affect residential mobility through income? The underlying premise is that higher educational attainment leads to greater income opportunities, which in turn may enable or constrain an individual's ability to move. By examining this potential mediating relationship, the research aims to uncover how social and economic factors are interconnected in influencing where people live and how frequently they relocate. This inquiry is particularly relevant in understanding patterns of socioeconomic mobility, housing stability, and spatial inequality in the United States.

4. Research Study

4.1. Dataset

The analysis utilizes data from the 2023 U.S. Census American Community Survey (ACS) 1-Year Public Use Microdata Sample (PUMS) Dataset. This dataset provides detailed population and housing information collected by the U.S. Census Bureau, offering a rich resource for social and economic analysis. The data was accessed through the official ACS microdata portal (https://www.census.gov/programs-surveys/acs/microdata/access.html) and further supported by variable documentation available at the Census Bureau's API directory (https://api.census.gov/data/2023/acs/acs5/pumspr/variables.html). For the purposes of this study, a random sample of 50,000 records was extracted from the larger dataset to ensure manageable data handling while maintaining representativeness. The sample provides a robust foundation for statistical analysis, enabling insights into the relationships among educational attainment, household income, and residential mobility across the United States.

4.2. Statistical Model

The statistical model employed in this research follows a sequential pathway: Educational Attainment (SCHL) → Household Income (HINCP) → Mobility Status (MIG). This model is designed to explore the relationships between educational attainment, household income, and residential mobility.

The model begins with SCHL, which represents an individual's level of educational attainment. Education is widely recognized as a fundamental determinant of socioeconomic status, and in this analysis, it serves as the primary independent variable. It is assumed that higher levels of education increase individuals' access to better employment opportunities, which in turn can lead to higher earnings.

The second variable in the model is HINCP, or household income in the past 12 months. This variable is treated as a mediating factor that is influenced by educational attainment. The underlying assumption is that educational attainment impacts income levels—those with higher educational credentials are more likely to secure well-paying jobs and thus have higher household incomes.

Finally, the model includes MIG, which refers to mobility status—specifically, whether individuals have changed their residence within the past year. This variable is considered the outcome of interest in the model. The hypothesis is that household income may influence an individual's or household's ability or need to move. For example, higher-income households may have more flexibility and resources to relocate, while lower-income households may move out of necessity or due to housing instability.

By structuring the model as SCHL → HINCP → MIG, the analysis seeks to capture the indirect effect of educational attainment on mobility through income. This pathway allows for a deeper understanding of how education and income together shape patterns of residential stability and change.

Table 1. Variable Names and Labels

Variable	Description	Codes / Values
SCHL	Educational attainment	bb: N/A (less than 3 years old) 01: No schooling completed 02: Nursery school, preschool 03: Kindergarten 04–14: Grade 1 to Grade 11 15: 12th grade – no diploma 16: Regular high school diploma 17: GED or alternative credential 18: Some college, less than 1 year 19: 1+ years of college, no degree 20: Associate's degree 21: Bachelor's degree 22: Master's degree 23: Professional degree beyond bachelor's 24: Doctorate degree
HINCP	Household income (past 12 months; adjust with ADJINC)	bbbbbbb: N/A (GQ/vacant) 0: No household income

		-59999: Loss of -$59,999 or more -59998 to -1: Loss of $1 to -$59,998 1 to 9,999,999: Total household income in dollars
MIG	Mobility status (lived in same house 1 year ago)	b: N/A (less than 1 year old) 1: Yes, same house (nonmovers) 2: No, outside US and Puerto Rico 3: No, different house in US or Puerto Rico

5. Demonstration on Software and the Results

5.1. Mediation Analysis using continuous variables

In this mediation analysis, we examined the relationship between educational attainment and residential mobility, and whether this relationship is mediated by household income, using all variables in their continuous forms. Educational attainment was measured on a progressive scale, ranging from no formal schooling to advanced graduate degrees. Household income was treated as a continuous variable, representing annual income in dollars. Residential mobility, the outcome of interest, was coded as a binary variable indicating whether an individual had changed residences in the past year.

In the study, we used a resampling technique called bootstrapping, in both the continuous and binned mediation analyses because it provides a robust, assumption-free method for assessing the statistical significance and precision of mediation effects. It is particularly suited for mediation because it handles the non-normality of indirect effects naturally and provides more accurate inference than classical parametric methods like the Sobel test.

Bootstrapping is a resampling technique used to estimate the sampling distribution of a statistic by repeatedly drawing samples—with replacement—from the observed data. It allows researchers to derive confidence intervals and p-values without relying on strict parametric assumptions about the distribution of the data, such as normality or homoscedasticity.

In the context of mediation analysis, particularly when estimating the indirect effect (i.e., the product of two regression coefficients: path *a* and path *b*), the sampling distribution of this product is often non-normal and asymmetric. Traditional parametric methods, such as the Sobel test, assume a normal distribution for the product of coefficients, which can lead to inaccurate or biased inference, especially when the indirect effect is small or the sample size is moderate.

To address this, we applied nonparametric bootstrapping in both the continuous and binned mediation analyses (using the mediate() function with boot = TRUE and sims = 1000). This process involved: repeatedly (1,000 times) resampling from the original dataset, recalculating the mediation model in each resample, generating a distribution of the Average Causal Mediation Effect (ACME), Average Direct Effect (ADE), and Proportion Mediated across all resamples, and constructing percentile-based confidence intervals for these estimates from the empirical distribution.

The primary reason for using bootstrapping here is that the product of two normally distributed variables (a × b) is not normally distributed. This makes traditional inference methods unreliable for testing the significance of the indirect effect. Bootstrapping avoids this issue by empirically generating the distribution, allowing for accurate estimation of confidence intervals and p-values, even with complex or skewed data structures.

Table 2: Mediation Analysis Using Continuous Variables

Effect	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	-0.000658	-0.000796	0.000000	<2e-16
ACME (treated)	-0.000649	-0.000785	0.000000	<2e-16
ADE (control)	-0.003124	-0.003922	0.000000	<2e-16
ADE (treated)	-0.003115	-0.003910	0.000000	<2e-16
Total Effect	-0.003773	-0.004596	0.000000	<2e-16
Prop. Mediated (control)	0.174337	0.135218	0.220000	<2e-16
Prop. Mediated (treated)	0.171902	0.133207	0.220000	<2e-16
ACME (average)	-0.000653	-0.000791	0.000000	<2e-16
ADE (average)	-0.003120	-0.003916	0.000000	<2e-16
Prop. Mediated (average)	0.173119	0.134213	0.220000	<2e-16

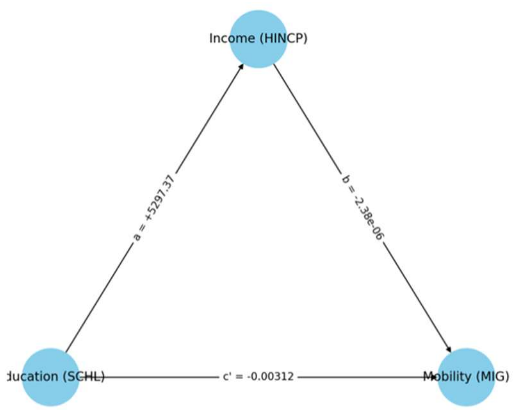
The results of the mediation analysis revealed a significant indirect effect, indicating that household income partially mediates the relationship between education and mobility. The Average Causal Mediation Effect (ACME)—the portion of education's influence that operates through income—was estimated at -0.000652, with a 95% confidence interval that excluded zero and a p-value of less than 0.001. Though the magnitude appears small, this effect is statistically robust, especially given the large sample size (n = 50,000). This negative value means that as individuals attain higher levels of education, their income increases, which is in turn associated with a lower likelihood of moving.

In addition to the mediated path, the Average Direct Effect (ADE) was also statistically significant, with an estimate of -0.00311. This suggests that even after accounting for income, education still has a direct influence on mobility. In other words, education contributes to residential stability in ways that go beyond income, possibly through increased job security, access to stable housing, or stronger social ties. Individuals with higher education may be more likely to own their homes, have more predictable careers, or live in more desirable areas—all of which could reduce the likelihood of moving.

The Total Effect of education on mobility, which combines both the direct and indirect components, was estimated at -0.00377 and was also highly significant. This confirms that, overall, higher education is associated with decreased residential mobility.

Of particular interest is the proportion mediated, which was approximately 17%. This tells us that income accounts for about one-sixth of the total effect of education on mobility. While this is a meaningful share, it also reveals that the majority—around 83%—of education's impact on mobility is not explained by income alone. This finding underscores the multifaceted value of education in shaping life outcomes. Education appears to grant individuals not only economic benefits but also social and structural advantages that contribute to residential stability.

Figure 1. Path Diagram: Mediation Analysis Using Continuous Variables



The graphical summary of the mediation analysis helps visualize these relationships. The ACME and ADE estimates are both negative and clearly fall below zero, with tight confidence intervals that indicate precision. The total effect is also negative and distinct from zero. This consistent direction across all effects reinforces the conclusion that higher education is associated with lower mobility, and that this is due in part—but not entirely—to the financial benefits that education brings.

The continuous mediation analysis paints a rich picture of how education stabilizes residential behavior. While income is a significant mediator, it accounts for only a portion of the relationship. Education appears to offer other non-monetary forms of security—be it career predictability, community engagement, or access to home ownership—that also reduce mobility. This underscores the value of education not just as a path to higher earnings, but as a foundation for a more stable, anchored life. In an age of increasing mobility and economic uncertainty, these findings affirm the deep, stabilizing power of education across both financial and social dimensions.

5.2. Mediation Analysis using binned variables

In the second mediation analysis, we explored the relationship between educational attainment and residential mobility using binned variables. Both education and income were grouped into discrete categories, representing meaningful real-world thresholds such as educational degrees and income brackets. This approach allows us to examine how moving from one categorical level to another—such as from high school to college, or from low to middle income—affects the likelihood of changing one's residence, and whether that relationship is mediated by household income.

The findings of this analysis reveal a strong and statistically significant mediation effect. The Average Causal Mediation Effect (ACME), which quantifies the portion of education's impact on mobility that operates through income, was estimated at -0.00767. This negative and highly significant result ($p < 2e-16$) indicates that individuals with higher educational attainment tend to fall into higher income categories, and this in turn is associated with a lower likelihood of residential mobility. The narrow confidence interval confirms the precision of this estimate.

In contrast, the Average Direct Effect (ADE), which represents the influence of education on mobility not explained by income, was not statistically significant ($p = 0.16$). Its confidence interval included zero, suggesting that once income is accounted for, education on its own does not meaningfully influence residential mobility. This indicates that the stabilizing effects of education on residential behavior may largely operate through the economic advantages it confers.

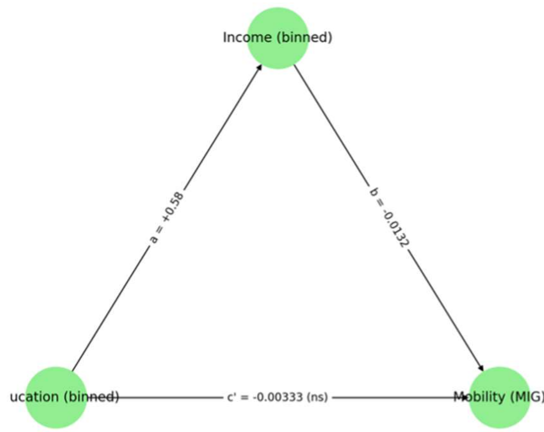
The Total Effect, combining both direct and indirect pathways, was estimated at -0.01101 and was statistically significant. This means that education overall is associated with reduced mobility, but this effect is primarily indirect, mediated through higher income. Most notably, the proportion mediated—the percentage of the total effect explained by the mediation path—was nearly 70%. This suggests that income accounts for the majority of education's influence on mobility in this model.

Table 3: Mediation Analysis Using Binned Variables (Bootstrap, 1,000 Simulations)

Effect	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	-0.00775	-0.00863	-0.01000	<2e-16
ACME (treated)	-0.00759	-0.00839	-0.01000	<2e-16
ADE (control)	-0.00341	-0.00828	0.00000	0.15
ADE (treated)	-0.00325	-0.00793	0.00000	0.15
Total Effect	-0.01100	-0.01613	-0.01000	<2e-16
Prop. Mediated (control)	0.70433	0.50184	1.17000	<2e-16
Prop. Mediated (treated)	0.69012	0.47932	1.17000	<2e-16
ACME (average)	-0.00767	-0.00850	-0.01000	<2e-16

ADE (average)	-0.00333	-0.00811	0.00000	0.15
Prop. Mediated (average)	0.69722	0.49058	1.17000	<2e-16

Figure 2. Path Diagram: Mediation Analysis Using Binned Variables



The accompanying plot visually reinforces these findings. The ACME point estimate lies well below zero with tight confidence bounds, indicating strong and significant mediation. The ADE, on the other hand, crosses the zero line, visually confirming its lack of statistical significance. The total effect is clearly negative and significant, as expected.

The binned mediation analysis presents a compelling narrative: education reduces residential mobility primarily by increasing household income. The strong indirect effect and lack of a direct effect suggest that once income level is accounted for, education no longer has a standalone influence on whether individuals move. Compared to the continuous model, the binned approach reveals a more substantial and clearer mediation pathway, emphasizing the role of categorical shifts over gradual change. Together, both analyses provide a richer understanding of how education shapes life trajectories, pointing to income as the key mechanism linking educational attainment to residential decisions.

5.3. Sobel tests

The Sobel test is a classic way to test the significance of the indirect (mediation) effect. However, it assumes normality of the indirect effect, so it's less robust than bootstrapping, but it's still useful as a quick check.

The Sobel test provides a parametric test of the significance of the indirect effect. We manually computed the Sobel z-test (using regression outputs) for both the continuous model using **HINCP** as mediator, and the binned model using **HINCP** with numeric encoding as mediator.

The Sobel z-value represents how many standard deviations away the indirect effect (a × b) is from zero. A larger absolute z-value (typically > z-critic = 1.96) indicates the indirect effect is statistically significant at the 0.05 level.

Sobel Test for Continuous Variables:

- **Sobel z** = -12.42 → The test statistic is larger than 1.96, hence the indirect effect is statistically significant at the 0.05 level.
- **p = 0** → Since p-value < 0.05, the indirect effect is statistically significant at 0.05 level.

Sobel Test for Binned Variables

- **Sobel z** = -20.62 → The test statistic is larger than 1.96, hence the indirect effect is statistically significant at the 0.05 level.
- **p = 0** → Since p-value < 0.05, the indirect effect is statistically significant at 0.05 level.

In conclusion, there is sufficient evidence of a significant indirect effect of schooling (SCHL) on migration status (MIG) through household income (HINCP), suggesting that education influences migration because it affects income. This supports the mediation hypothesis, as it also supports the mediation analysis results through the non-parametric bootstrapping method tested previously.

6. Discussion

7. Conclusion

The continuous mediation analysis paints a rich picture of how education stabilizes residential behavior. While income is a significant mediator, it accounts for only a portion of the relationship. Education appears to offer other non-monetary forms of security—be it career predictability, community engagement, or access to homeownership—that also reduce mobility.

This underscores the value of education not just as a path to higher earnings but as a foundation for a more stable, anchored life. In an age of increasing mobility and economic uncertainty, these findings affirm the deep, stabilizing power of education across both financial and social dimensions. The findings of the binned analysis reveal a strong and statistically significant mediation effect. Most notably, the proportion mediated—the percentage of the total effect explained by the mediation path—was nearly 70%. This suggests that income accounts for the majority of education's influence on mobility in this model. Further research could explore moderation effects, such as how the relationship between education, income, and mobility might differ based on sex or other demographic variables. Such analyses could reveal nuanced patterns and inform more targeted policies.